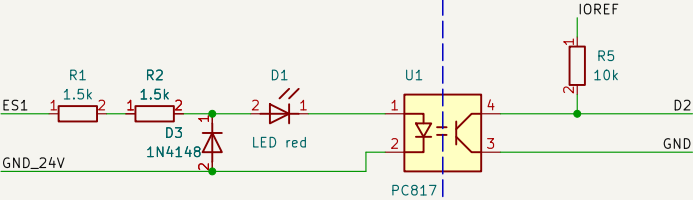


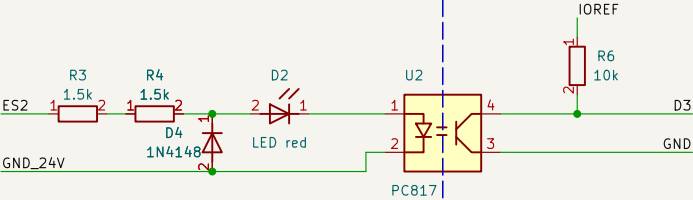
Two e-stop inputs, 24 V loop through PC817 – shield for Arduino UNO R4 WiFi / UNO Q

Loop current = $(24\text{ V} - 1.2\text{ V opto} - 2.0\text{ V LED}) / 3.0\text{ k} = 6.9\text{ mA}$. Each 1.5 k (1/4 W): 72 mW at 24 V, 109 mW at 28.8 V.
Button pressed, wire cut or 24 V lost -> no current -> pin HIGH = STOP. F1 protects the 24 V wires going out to the buttons.
GND_24V and GND are NOT connected: this is the isolation barrier.
Pull-ups go to IOREF: 5 V on the UNO R4 WiFi, 3.3 V on the UNO Q. Never to the 5V pin. Sketch: pinMode(D2/D3, INPUT).

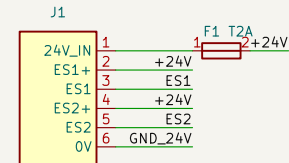
E-stop 1: button released (contact closed) -> D2 LOW, LED lit



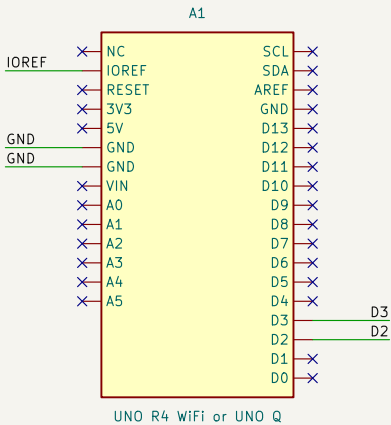
E-stop 2: button released (contact closed) -> D3 LOW, LED lit



ISOLATION



24V + e-stops



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Sheet: /
File: pc817_bench.kicad_sch

Title: E-stop inputs: 24 V loop through PC817

Size: A4
KiCad E.D.A. 10.0.6

Date: 2026-10-02

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